S&F, Ch.2
Starykh

Solutions HW03

P $\boxed{2.1}$ $0.5\,h \cdot 80\,\frac{km}{h} = 40\,km$

$\frac{1}{5}\,h \cdot 100\,\frac{km}{h} = 20\,km$

$45\,min \cdot 40\,\frac{km}{h} = \frac{3}{4}\,h \cdot 40\,\frac{km}{h} = 30\,km$

$15\,min = \frac{1}{4}\,h$ lunch

(a) Average speed $= \frac{\text{distance}}{\text{time}} = \frac{40+20+30}{0.5+0.2+0.75+0.25}\,\frac{km}{h} = \frac{90\,km}{1.7\,h} = 52.9\,\frac{km}{h}$

(b) $90.0\,km$

P $\boxed{2.3}$ Boat A: $t = 2\,hr$

Boat B: $2\,h + \frac{60\,km}{90\,km/h} = \left(2 + \frac{2}{3}\right)hr = 2\,hr\ 40\,min$

At $t = 2\,hr$ boat B is at the opposite bank of the lake, 60 km away

(a) Boat A wins by $\boxed{40\,min}$, or 60 km.

(b) Average velocity $\bar{v} = \frac{\Delta x}{\Delta t} = \frac{0\,km}{2\,hr} = \boxed{0\,km/h}$

also calculate average speed for each boat

Average speed: boat A $\bar{v}_A = \frac{60+60}{2}\,\frac{km}{h} = 60\,km/h$

boat B $\bar{v}_B = \frac{60+60\,km}{2\frac{2}{3}\,h} = \frac{120\,km}{\frac{8}{3}\,hr} = \frac{360}{8}\,\frac{km}{h} =$

$= 45\,km/h$

P $\boxed{2.5}$ $35.0\,min \cdot 85.0\,\frac{km}{h} = \frac{35}{60} \cdot 85\,km = 49.6\,km$

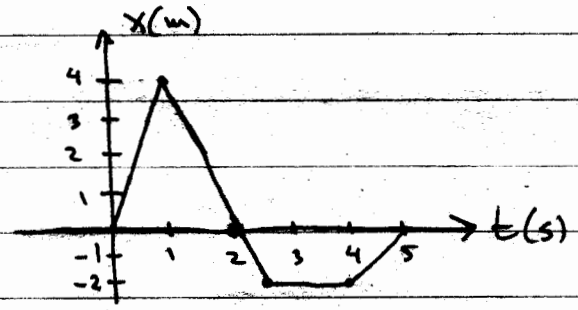
stops for $15\,min = 0.25\,h$

then continues north, 130 km in 2 hr

(a) Total displacement $= 130 + 49.6\,km = 179.6\,km = \boxed{180\,km}$

(b) Average velocity $= \frac{179.6\,km}{\left(\frac{35}{60} + 0.25 + 2\right)hr} = \boxed{63.4\,km/h}$

2.7



Denote $x(t)$ position at time t.

Find average velocity $\bar{v}$ in intervals

(a) 0 to 1.0s:

$$\bar{v} = \frac{x(1.0)-x(0)}{1.0\,s} = \frac{4-0}{1.0}\,\frac{m}{s} = 4.0\,m/s$$

(b) 0 to 4.0s:

$$\bar{v} = \frac{x(4.0)-x(0)}{4.0\,s} = \frac{-2-0}{4.0}\,\frac{m}{s} = -0.5\,m/s$$

(c) 1.0s to 5.0s: $\bar{v} = \dfrac{x(5.0)-x(1.0)}{(5-1)s} = \dfrac{0-4}{4}\,\dfrac{m}{s} = -1.0\,m/s$

(d) 0 to 5.0s: $\bar{v} = \dfrac{x(5.0)-x(0)}{5.0\,s} = \dfrac{0}{5}\,\dfrac{m}{s} = 0\,m/s$

2.9

(a) forward motion: graphs a, e, f.

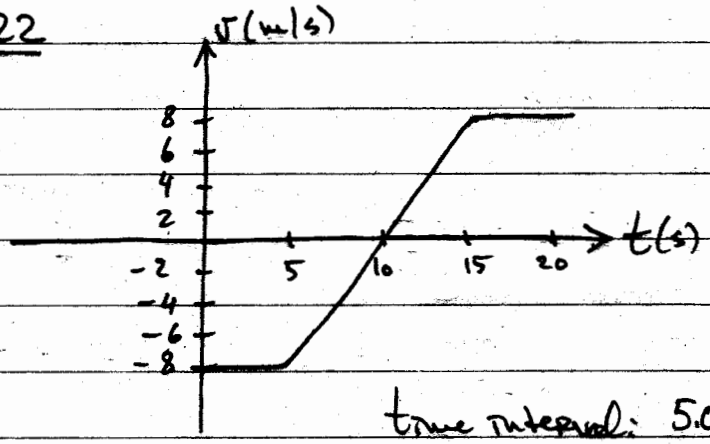
(b) backward motion: graph c

(c) constant velocity: graphs f and d.

(d) greatest constant velocity: graph f.

(e) no movement: graph d.

2.22



(a) average acceleration

$$\bar{a} = \frac{\Delta v}{\Delta t}$$

time interval: 0 to 5.0s

$$\bar{a} = \frac{v(5.0)-v(0)}{5.0\,s} = \frac{-8-(-8)}{5.0}\,\frac{m}{s^2} = 0\,m/s^2$$

time interval: 5.0s to 15s

$$\bar{a} = \frac{v(15)-v(5)}{(15-5)s} = \frac{8-(-8)}{10}\,\frac{m}{s^2} = 1.6\,m/s^2$$

0 to 20s: $\bar{a} = \dfrac{v(20)-v(0)}{20-0} = \dfrac{8-(-8)}{20}\,\dfrac{m}{s^2} = 0.8\,m/s^2$

(b) Instantaneous acceleration $a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}$

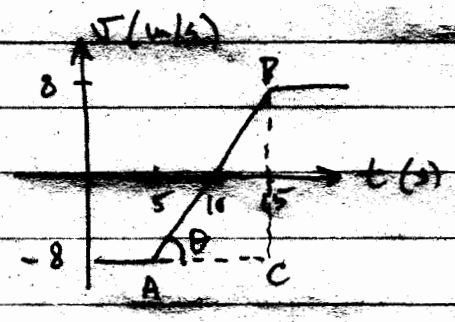
is the slope of the tangent line to $v(t)$ curve.

At $t = 2.0\,s$ and $t = 18\,s$ the tangent lines are horizontal

$\Rightarrow$ the slope $= 0$ $\Rightarrow$ acceleration $a = 0\ m/s^2$ at these times.

At $t = 10\,s$ the tangent line coincides with the curve

(which is a straight line for $5 \le t \le 15\,s$)



Find slope of line AB; this slope is given by $\tan(\theta)$ in the triangle ABC

Side $|AC| = 15 - 5 = 10\,s$

Side $|BC| = 8 - (-8) = 16\ m/s$

$$\Rightarrow \tan(\theta) = \frac{|BC|}{|AC|} = \frac{16\ m/s}{10\ s} = 1.6\ m/s^2$$

$\Rightarrow$ acceleration at $t = 10\,s$ is $a = 1.6\ m/s^2$.